# Chess Card PCB — Specification v2

A slightly oversized business card PCB (95 × 60 mm) that runs a dual-encoder energy-based chess AI on an ESP32-S3. The human plays one side; the card plays the other. Charges via USB-C or trickle-charges from a small back-mounted solar cell. No wireless connectivity required during play.

Designed for JLCPCB fabrication with PCBA assembly service.

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## 1. Physical Specifications

| Property | Value |
|---|---|
| Outline dimensions | 95 × 60 mm |
| Corner radius | 2 mm rounded |
| PCB thickness | 0.8 mm |
| Layers | 2 (top + bottom), standard FR4 |
| Soldermask | Black |
| Silkscreen | White, both sides |
| Surface finish | ENIG (gold) — premium look, good for exposed touch pads |
| Special feature | 0.4 mm depth-controlled milling pocket on back (LiPo recess) |
| Assembly | JLCPCB PCBA for all SMD components |

**Post-PCBA hand assembly:** LiPo battery wires (to SMD JST-PH connector), coin haptic motor (2 wires), solar cell (2 wires), e-paper panel (FPC, no soldering — slides into latching connector). No through-hole soldering required.

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## 2. Front Face Layout

```
┌──────────────────────────────────────────────────────────┐
│                                                          │
│  ┌──────────────────────────────────────┐   [ ▲ UP   ] │
│  │                                      │              │
│  │      2.9" E-Paper Display            │   [ ● SEL  ] │
│  │      (296 × 152px, 4-grayscale)      │              │
│  │                                      │   [ ▼ DN   ] │
│  │                                      │              │
│  └──────────────────────────────────────┘   [ USB-C ]  │
│                                                          │
│          Neelay Ranjan | 773-668-0069 | rneelay@gmail.com  │
└──────────────────────────────────────────────────────────┘
```

- **E-paper:** left-aligned, top-aligned, ~70 × 38 mm active area
- **Touch column:** right side, vertically centered (3 pads in a column)
- **USB-C:** right edge, bottom right corner
- **Branding silkscreen:** bottom strip of front face, white text on black soldermask
  - Text height: ~1.2 mm, clean sans-serif
  - Centered horizontally across full card width
  - Small chess piece silhouette or board corner motif

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## 3. Back Face Layout

```
┌──────────────────────────────────────────────────────────┐
│                                                          │
│      [ESP32-S3-WROOM-2 Module]    [Solar Cell, 40×30mm]  │
│      (antenna toward top edge)                           │
│                                                          │
│      [BQ25570] [MCP73832] [XC6220] [MAX17048] [Motor pads]  │
│                                                          │
│      [500mAh LiPo — sits in 0.4mm recess]                │
│                                                          │
│      [● ● ● ● ●] ← 5 charge level LEDs                   │
│                                                          │
│      [RST bridge pads]              [Reset/restart pads] │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

- **ESP32-S3 module:** upper center, antenna oriented toward top edge
- **Solar cell:** upper right, flush adhesive mount on back
- **Power management ICs:** middle horizontal strip
- **LiPo battery:** lower middle, sits in milled pocket, JST-PH SMD connector
- **5 charge LEDs:** below LiPo, visible row of 0402 LEDs
- **Reset bridge pads:** lower edge, 2 small exposed copper pads to short for hard reset

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## 4. ESP32-S3 Module

| Property | Value |
|---|---|
| Part | ESP32-S3-WROOM-2-N16R8 |
| Specs | Dual-core Xtensa LX7 @ 240MHz, 8MB PSRAM, 16MB Flash |
| Package | Castellated module, 18 × 25 × 3.1 mm |
| Placement | Back of card, upper center |
| Antenna orientation | Pointed toward top edge of card |
| LCSC search | "ESP32-S3-WROOM-2-N16R8" |

**Layout requirements:**

- **Antenna keepout: 15mm clearance zone** at top edge of module — no copper pours, traces, or vias whatsoever in this zone
- **Decoupling:** 100nF + 10µF ceramic caps on every VDD pin, placed within 0.5 mm
- **Bulk cap:** 100µF on 3.3V rail near module VDD pins (handles inference current spikes)

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## 5. E-Paper Display

| Property | Value |
|---|---|
| Recommended part | GoodDisplay GDEY029F51 |
| Size | 2.9" diagonal |
| Resolution | 296 × 152 pixels |
| Grayscale | 4 levels (white, light gray, dark gray, black) |
| Active area | ~67 × 35 mm |
| Refresh modes | Full refresh (~2s, 4-grayscale, flashy) + fast refresh (~300ms, B/W, no flash) |
| Idle power | ~0 µA (bistable — holds image with no power) |
| Voltage | 3.3V VCI |
| Connector | 24-pin FPC, 0.5 mm pitch, top-contact (verify against datasheet) |
| Placement | Front of card, flush to top-left corner |

**Refresh strategy during gameplay:**

- Use **fast refresh** for all in-game updates (cursor movement, AI move display, captured pieces)
- Run **full refresh once between games** to clear ghosting

**Required cap network:** The internal charge pump needs ~5 dedicated 1µF ceramic caps on VPOS, VNEG, VCOM, VGH/VSH, VGL/VSL pins. **Pull the GoodDisplay reference schematic for GDEY029F51 specifically and replicate the cap network exactly.** This is the #1 cause of e-paper boot failures.

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## 6. Capacitive Touch Buttons

Three buttons in a vertical column on the right side of the front face. Uses ESP32-S3 built-in capacitive touch — no external IC needed.

```
Top of card:
                [ ▲ UP ]      ← GPIO4

                [ ● SEL ]     ← GPIO5  (RTC touch wake source)

                [ ▼ DN ]      ← GPIO6
Bottom of card:
```

| Pad | GPIO | Function |
|---|---|---|
| UP | GPIO4 | Cycle move list up |
| SELECT | GPIO5 | Confirm move / pick-and-place / wake from sleep |
| DOWN | GPIO6 | Cycle move list down |

**Pad design:**

- Diameter: 8 mm exposed copper circles, front layer
- Pitch: 14 mm vertical spacing
- Surface finish: ENIG over pads (clean, finger-friendly)
- Soldermask: removed over pad area
- Silkscreen labels (▲, ●, ▼) on white silkscreen adjacent to each pad

**Layout rules:**

- **No ground pour under or within 5 mm of any touch pad** (kills capacitive sensitivity)
- Hatched ground only (not solid) within 10 mm of pads if ground reference needed
- Route touch traces on front layer only, short as possible, away from SPI traces and motor wires
- All three trace lengths roughly matched (within 5 mm of each other) for consistent sensitivity

**Wake behavior:**

- Only the **SELECT pad (GPIO5)** is monitored by the RTC touch peripheral during deep sleep
- Touch wake adds ~10 μA to deep sleep current
- Firmware should dynamically calibrate touch threshold on each wake (read baseline for 50 ms, set threshold relative to baseline) to compensate for temperature/humidity drift
- Wake event must persist for 100 ms before counting (noise rejection)

## 7. ERM Haptic Motor

| Property | Value |
|---|---|
| Motor | NFP-C0820B or Jinlong C0820B02F (8 mm × 2.0 mm coin) |
| Drive transistor | 2N7002 N-channel MOSFET (SOT-23) |
| Flyback protection | 1N4148W diode across motor terminals (cathode to + terminal) |
| Gate driver pin | GPIO13 |
| Power | LiPo+ direct (NOT 3.3V rail — avoids rail dipping during inference) |
| Mounting | Adhered to back of card with double-sided tape, post-assembly |
| PCB interface | Two solder pads for motor lead wires |

**Important:** Motor draws 60–80 mA spikes. Driving from LiPo+ instead of 3.3V prevents brown-outs of e-paper SPI and touch pads during haptic events.

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## 8. Power Management

### 8.1 USB-C Charging Path

| Component | Detail |
|---|---|
| USB-C connector | 16-pin SMD (LCSC: "USB-C 16pin SMD") |
| CC1/CC2 | 5.1kΩ pull-down resistors to GND (signals USB device, enables 5V) |
| ESD protection | USBLC6-2SC6 on D+/D- lines |
| D+ → ESP32-S3 | GPIO20 |
| D- → ESP32-S3 | GPIO19 |
| VBUS → | MCP73832 input (via Schottky diode) |

**Programming:** Native USB-CDC over the S3's hardware USB peripheral. No CH340/CP2102 bridge IC needed. Flash via `idf.py flash` over USB-C.

### 8.2 Solar Trickle Charging Path

| Component | Detail |
|---|---|
| Solar cell | 40 × 30 mm amorphous, 5V, ~30 mW peak |
| Mounting | Adhered to top of back face, flush, near top edge |
| Harvester IC | BQ25570 (TI, ultra-low-power) |
| Package | QFN-20 |
| Quiescent current | 330 nA |
| Output | Charges LiPo directly, programmed for 4.2V max |
| External components | 22μH inductor, input/output caps, MPPT-setting resistors |
| LCSC search | "BQ25570RGRR" |

**Brownout protection:** BQ25570 VBAT_OK output → XC6220 EN pin (with small RC filter). When LiPo voltage drops below safe threshold, LDO disables and ESP32 powers off cleanly. Prevents brown-out corruption of RTC memory and flash.

### 8.3 LiPo Charger

| Component | Detail |
|---|---|
| Charger IC | MCP73832T-2ACI/OT |
| Package | SOT-23-5 |
| Charge current | 500 mA (fast charge) |
| PROG resistor | 2.0 kΩ (sets 500 mA) |
| Input source | USB-C VBUS (BQ25570 charges LiPo independently from solar side) |
| Status output | STAT pin → ESP32 (via voltage divider, optional) |
| LCSC search | "MCP73832" |

### 8.4 Schottky OR-ing

| Component | Detail |
|---|---|
| Part | BAT54S (dual Schottky in SOT-23) |
| Purpose | Prevents BQ25570 and MCP73832 from back-feeding each other |
| Diode 1 | BQ25570 VBAT output → LiPo+ |
| Diode 2 | MCP73832 VBAT output → LiPo+ |

## 8.5 3.3V LDO

| Component | Detail |
|---|---|
| LDO | XC6220B331MR-G |
| Package | SOT-23-5 |
| Output | 3.3V fixed, 1A max |
| Input | LiPo 3.0–4.2V |
| EN pin | Tied to BQ25570 VBAT_OK (brownout protection) |
| Decoupling | 1µF ceramic input + 1µF ceramic output |
| Powers | ESP32-S3, e-paper, touch pads, fuel gauge, status LEDs |

## 8.6 Fuel Gauge

| Component | Detail |
|---|---|
| Part | MAX17048G+T10 |
| Package | SOT-23-6 |
| Function | Reports battery percentage and voltage over I2C |
| I2C address | 0x36 |
| SDA → ESP32-S3 | GPIO38 |
| SCL → ESP32-S3 | GPIO39 |
| Decoupling | 1µF ceramic on VCC |
| LCSC search | "MAX17048" |

### 8.7 Charge Level LEDs

Five 0402 LEDs in a row on the back of the card, below the LiPo. Driven by ESP32 GPIOs based on MAX17048 reading.

| LED | GPIO | Color | Threshold (% charge) |
|---|---|---|---|
| LED1 | GPIO14 | Red | >0% |
| LED2 | GPIO15 | Red | >20% |
| LED3 | GPIO16 | Yellow | >40% |
| LED4 | GPIO17 | Green | >60% |
| LED5 | GPIO18 | Green | >80% |

**Behavior:**

- Solid bars when card is awake or just woken
- Walking pattern when actively charging
- Off during deep sleep
- Each LED in series with 1kΩ resistor to GPIO

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## 9. Battery

| Property | Value |
|---|---|
| Recommended cell | EEMB LP503035 |
| Capacity | 500 mAh |
| Dimensions | 35 × 30 × 5 mm |
| Voltage | 3.7V nominal, 4.2V max |
| Protection | Built-in PCM (overcharge / overdischarge / short circuit) |
| Connector | JST-PH 2.0 mm 2-pin SMD (S2B-PH-SM4-TB), JLCPCB-placeable |
| Mounting | Sits in 0.4 mm milled pocket on back of card |

**Final thickness at battery location:** 0.4 mm (remaining PCB) + 5 mm (LiPo) = 5.4 mm. This is the thickest point on the card. ESP32-S3 module elsewhere is 0.8 + 3.1 = 3.9 mm.

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## 10. Reset Bridge Pads

In place of a tactile reset button. Two small exposed copper pads on the back of the card, near bottom edge.

| Spec | Value |
|---|---|
| Pad size | 1.5 × 1.5 mm each |
| Pad spacing | 1.5 mm apart |
| Function | Bridging pads shorts EN to GND, hard-resets ESP32 |
| Silkscreen label | "RST" |
| Field finish | Cover with hot glue or Kapton tape after firmware loading and validation |

**No power switch** — card is always in deep sleep when not in use, woken by SELECT touch. The reset bridge pads are for emergency use during development and rare in-field resets.

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## 11. Power Architecture Diagram

```
USB-C VBUS ──────[BAT54S]──────┐
                               ├──> MCP73832 VDD_IN ──> LiPo (+)
Solar Cell ──> BQ25570 ────────┘                        |
                  |                                     |
                  └──> VBAT_OK ──> XC6220 EN            |
                                                        |
LiPo (+) ──> XC6220 IN ──> 3.3V rail                   |
LiPo (-) ──> GND                                        |
LiPo (+) ──> MAX17048 (battery monitor I2C)             |
LiPo (+) ──> 2N7002 source ──> ERM motor return         |

XC6220 OUT ──> 3.3V rail ──> ESP32-S3 (3V3)
                           ──> E-paper VCC
                           ──> Touch pads (via GPIO)
                           ──> MAX17048 VCC
                           ──> Charge LEDs (via GPIO)

Solar tab (+) ──> BQ25570 VIN_DC
Solar tab (-) ──> GND

USB-C CC1 ──> 5.1kΩ ──> GND
USB-C CC2 ──> 5.1kΩ ──> GND
USB-C D+ ──> USBLC6 ──> ESP32-S3 GPIO20
USB-C D- ──> USBLC6 ──> ESP32-S3 GPIO19

ESP32-S3 GPIO11 ──> E-paper MOSI
ESP32-S3 GPIO12 ──> E-paper SCK
ESP32-S3 GPIO10 ──> E-paper CS
ESP32-S3 GPIO9  ──> E-paper DC
ESP32-S3 GPIO8  ──> E-paper RST
ESP32-S3 GPIO7  ──> E-paper BUSY

ESP32-S3 GPIO4  ──> Touch UP
ESP32-S3 GPIO5  ──> Touch SELECT (RTC wake source)
ESP32-S3 GPIO6  ──> Touch DOWN

ESP32-S3 GPIO13 ──> 2N7002 GATE ──> ERM motor
1N4148W ──> across ERM motor terminals (flyback)

ESP32-S3 GPIO14-18 ──> 5 charge level LEDs (via 1kΩ each)

ESP32-S3 GPIO38 ──> I2C SDA (MAX17048)
ESP32-S3 GPIO39 ──> I2C SCL (MAX17048)

ESP32-S3 EN ──> BQ25570 VBAT_OK (brownout protection)
ESP32-S3 EN ──> RST bridge pads ──> GND (manual reset)
```

```
ESP32-S3 EN ──> 10kΩ pull-up to 3.3V
```

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## 12. Pin Assignment Summary

| GPIO | Function | Notes |
|---|---|---|
| GPIO0 | (BOOT strap) | Leave with internal pull-up; not used |
| GPIO3 | (avoid — strapping) | JTAG source select |
| GPIO4 | Touch UP | Cap touch input |
| GPIO5 | Touch SELECT | Cap touch input + RTC wake |
| GPIO6 | Touch DOWN | Cap touch input |
| GPIO7 | E-paper BUSY | Input |
| GPIO8 | E-paper RST | Output |
| GPIO9 | E-paper DC | Output |
| GPIO10 | E-paper CS | Output |
| GPIO11 | E-paper MOSI | SPI output |
| GPIO12 | E-paper SCK | SPI clock |
| GPIO13 | ERM motor gate | MOSFET drive |
| GPIO14 | Charge LED 1 | 0402 red |
| GPIO15 | Charge LED 2 | 0402 red |
| GPIO16 | Charge LED 3 | 0402 yellow |
| GPIO17 | Charge LED 4 | 0402 green |
| GPIO18 | Charge LED 5 | 0402 green |
| GPIO19 | USB D- | Native USB |
| GPIO20 | USB D+ | Native USB |
| GPIO35–37 | (internal — PSRAM) | Not exposed on N16R8 |
| GPIO38 | I2C SDA | MAX17048 |
| GPIO39 | I2C SCL | MAX17048 |
| GPIO45, 46 | (avoid — strapping) | VDD_SPI, ROM print |
| EN | Reset / brownout | RST bridge pads + BQ25570 VBAT_OK |

## 13. UI Layout (E-Paper)

```
┌──────────────────────────────────────────────────────────┐
│  ┌──────────┐  ┌──────────┐  ┌──────────────────┐        │
│  │          │  │ Captured │  │ Thinking...      │        │
│  │  8x8     │  │  ♟♟♟♞    │  │ ▓▓▓▓▓▓▓▓▓▓ 60%   │        │
│  │  board   │  │          │  │                  │        │
│  │          │  │  ♝♝♘♖    │  │ AI move: Nf3     │        │
│  │ 128×128  │  │          │  │                  │        │
│  │          │  │ Last:    │  │ Your moves:      │        │
│  │          │  │ ...e5    │  │ ▶ e5             │        │
│  │          │  │          │  │   d6             │        │
│  │          │  │          │  │   c5             │        │
│  │          │  │          │  │ [▲▼ scroll]      │        │
│  │          │  │          │  │ [● confirm]      │        │
│  └──────────┘  └──────────┘  └──────────────────┘        │
│  ~128 px       ~70 px         ~98 px                     │
└──────────────────────────────────────────────────────────┘
   board          middle          right
```

**Layout rationale:** Board on the left, status/controls on the right (closest to the button column for shortest reach). User's hand operating the right-edge buttons doesn't block the board.

**Display regions:**

- **Board (left, 128×128 px):** 8×8 chess board with 16×16 px squares, piece sprites in 4 grayscale levels
- **Captured pieces panel (middle, ~70 px wide):** Shows pieces taken by each side, plus last move in algebraic notation
- **Status/controls panel (right, ~98 px wide):** "Thinking..." progress bar during AI inference, AI's chosen move, scrollable list of legal moves for human player, current selection indicator

**Battery percentage** shown briefly on wake via the 5 back-side LEDs. E-paper does not display battery during gameplay (saves screen real estate).

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## 14. User Interaction Model

### Wake from sleep

- Card in deep sleep by default
- Touch SELECT pad → RTC touch wakes ESP32 → boots in ~200 ms

**Pre-game (difficulty selection)**

- E-paper displays: "Difficulty: ▶ Beginner / Intermediate / Expert"
- UP/DOWN scroll between options
- SELECT confirms, starts game

**During AI's turn**

- "Thinking..." progress bar fills as inference runs (target: ≤10s per move)
- Haptic buzz when move is chosen
- AI's move shown in algebraic notation, board updates with arrow indicator
- Card transitions to user's turn

**During user's turn**

- List of legal moves displayed, ranked by AI's evaluation
- UP/DOWN scroll the list, current selection shown with ▶ marker and on-board highlight
- SELECT confirms the chosen move, board updates

**Game restart**

- Hold UP + DOWN simultaneously for 3 seconds → restart confirmation prompt
- SELECT confirms restart, returns to difficulty selection

**Demo mode (always-on)**

- When no game is in progress, card sleeps
- E-paper displays: name, contact info, dynamic QR code (encodes vCard or portfolio URL)
- Refresh once on long-interval wake (every few days) to prevent ghosting
- Touch SELECT to enter game mode

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## 15. Bill of Materials

| Ref | Component | Part | Package | Qty |
|---|---|---|---|---|
| U1 | Processor + RAM + Flash | ESP32-S3-WROOM-2-N16R8 | Module | 1 |
| U2 | LiPo charger | MCP73832T-2ACI/OT | SOT-23-5 | 1 |
| U3 | Solar harvester | BQ25570RGRR | QFN-20 | 1 |
| U4 | LDO 3.3V | XC6220B331MR-G | SOT-23-5 | 1 |
| U5 | Fuel gauge | MAX17048G+T10 | SOT-23-6 | 1 |
| U6 | USB ESD protection | USBLC6-2SC6 | SOT-23-6 | 1 |
| Q1 | MOSFET (haptic) | 2N7002 | SOT-23 | 1 |
| D1 | Schottky OR diode (dual) | BAT54S | SOT-23 | 1 |
| D2 | Flyback diode | 1N4148W | SOD-123 | 1 |
| D3–D7 | Charge level LEDs (mixed colors) | 0402 LEDs | 0402 | 5 |
| L1 | Inductor | 22µH | 0603 | 1 |
| C1–C12 | Decoupling caps | 100nF | 0402 | 12 |
| C13–C18 | Bulk caps | 10µF | 0805 | 6 |
| C19 | Bulk cap | 100µF | 1210 | 1 |
| C20–C24 | E-paper charge pump caps | 1µF | 0603 | 5 |
| R1 | PROG resistor | 2.0kΩ | 0402 | 1 |
| R2 | EN pull-up | 10kΩ | 0402 | 1 |
| R3–R4 | USB CC resistors | 5.1kΩ | 0402 | 2 |
| R5–R9 | LED current limit | 1kΩ | 0402 | 5 |
| R10–R11 | I2C pull-ups | 4.7kΩ | 0402 | 2 |
| R12–R14 | BQ25570 MPPT/feedback | per datasheet | 0402 | 3 |
| J1 | USB-C connector | 16-pin SMD | SMD | 1 |

| | | | | |
|---|---|---|---|---|
| J2 | LiPo connector | JST-PH 2.0 mm SMD (S2B-PH-SM4-TB) | SMD | 1 |
| J3 | FPC connector | 24-pin 0.5 mm pitch | SMD | 1 |
| — | E-paper panel | GoodDisplay GDEY029F51 (296×152, 4-grayscale) | External | 1 |
| — | LiPo battery | EEMB LP503035 (500 mAh, 35×30×5 mm, PCM) | External | 1 |
| — | Solar cell | 40×30 mm, 5V, ~30 mW amorphous | External | 1 |
| — | ERM haptic motor | 8 mm × 2.0 mm coin (NFP-C0820B) | External | 1 |

**Removed from v1:** Qi receiver coil, BQ51013B Qi IC, SPV1040 (replaced by BQ25570), NFC sticker, dedicated BOOT/RESET tactile buttons.

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## 16. PCB Layout Rules

1. **ESP32-S3 antenna keepout: 15 mm zone** at antenna end — no copper, traces, or vias whatsoever, both layers
2. **Touch pad ground exclusion: no ground pour within 5 mm of any touch pad** — destroys capacitive sensitivity. Hatched ground only within 10 mm if needed
3. **Touch traces:** front copper layer only, as short as possible, lengths matched within 5 mm, route away from SPI traces and motor wires
4. **Decoupling cap placement:** within 0.5 mm of every VDD/GND pin pair on ESP32-S3 module
5. **Ground plane:** solid pour on both layers (except keepout zones), via stitch every 5 mm around board perimeter
6. **SPI routing (e-paper):** short, matched length, away from power traces and haptic motor wires
7. **Solar cell area:** keep ground pour but no signal traces under cell adhesive footprint — avoid voltage offsets from light-induced photocurrent in traces
8. **Haptic motor traces:** keep away from touch pad traces — motor switching injects noise. Drive from LiPo+ rail, not 3.3V rail
9. **NFC: N/A** — replaced by dynamic e-paper QR
10. **USB-C data lines:** route as 90Ω differential pair, length matched, place USBLC6 within 5 mm of connector
11. **Fuel gauge placement:** close to LiPo+ sense point, short trace
12. **Charge LED placement:** row of 5 LEDs in a clear row on back, below LiPo, visible when card is held back-up
13. **Reset bridge pads:** lower edge of back, away from high-current charging traces and ESP32-S3 module heat
14. **Depth-controlled milling pocket:** 0.4 mm deep, sized to LiPo footprint + 0.5 mm clearance, on back side, marked clearly in fab drawing

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## 17. Stack-Up & Thickness

| Element | Thickness |
|---|---|
| PCB FR4 | 0.8 mm |
| ESP32-S3 module (back) | 3.1 mm |
| LiPo cell (back, in 0.4 mm recess) | 5 mm cell − 0.4 mm recess = 4.6 mm net protrusion |
| E-paper panel (front) | 1.0 mm |
| Solar cell (back) | 0.5 mm |

**Total thickness at thickest point (LiPo region):** 0.4 mm (residual PCB) + 5.0 mm (LiPo) = **5.4 mm**

**Total thickness at ESP32 module:** 0.8 mm (PCB) + 3.1 mm (module) = **3.9 mm**

**Front face (e-paper):** 0.8 mm + 1.0 mm = **1.8 mm**

Intentionally chunky — thicker than a credit card, but pocketable.

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## 18. Charging & Runtime

### Charging time (from flat to full, ~500 mAh)

| Source | Power input | Time to full |
|---|---|---|
| USB-C (any 5V, 500 mA+ source) | ~500 mA charge | ~70–80 min |
| Solar (outdoor sun) | ~15–20 mA | ~25 hours |
| Solar (indoor desk light, 500 lux) | ~1–2 mA | Trickle / idle compensation only |

### Battery life

- **Average game energy:** ~720 J (60 moves × 12 J/move including inference, e-paper, idle)
- **Battery energy at full charge:** 500 mAh × 3.7 V = 6660 J usable
- **Estimated games per charge:** ~5–6 games

### Standby current (deep sleep with RTC touch wake)

- ESP32-S3 deep sleep: ~10 µA
- XC6220 LDO quiescent: ~8 µA
- BQ25570 quiescent: ~0.3 µA
- MAX17048 quiescent: ~3 µA
- LiPo PCM: ~3 µA
- **Total: ~24 µA = ~90 µW**

**Solar break-even**

- Indoor desk light harvest: ~1–2 mW (after BQ25570 efficiency)
- Standby draw: ~0.09 mW
- **Net positive by ~10–20× during desk display** — solar trickle keeps card topped up between games

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## 19. Manufacturing Notes

1. Order board as standard 2-layer PCB — no rigid-flex
2. Specs: FR4, 0.8 mm, black soldermask, white silkscreen, ENIG finish, 2 mm corner radius
3. Specify **0.4 mm depth-controlled milling pocket** on back side, sized for LiPo footprint
4. Upload Gerbers + BOM + CPL for JLCPCB PCBA assembly
5. Initial run: **5 boards** (validate v1 before scaling)
6. All SMD components assembled by JLCPCB PCBA, including USB-C, FPC connector, JST-PH SMD
7. Post-assembly hand work:
    - Solder LiPo wires to JST-PH (no SMD soldering — slip-fit connector)
    - Adhere Qi coil **(removed — skip this step)**
    - Adhere solar cell to back, solder 2 wires to PCB pads
    - Attach ERM motor with adhesive, solder 2 wires to PCB pads
    - Slide e-paper FPC into J3 connector, close latch
    - Cover RST bridge pads with hot glue or Kapton after first flash + validation
8. First programming: USB-C with ESP-IDF, ESP32-S3 board package
9. Partition scheme: bootloader + app + dedicated `model` partition (5–10 MB) + nvs + spiffs
10. Flash model file separately from firmware via `parttool.py` to avoid reflashing 10 MB on every code change

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## 20. Repository

- **Visibility:** Private
- **Structure:** schematic / PCB / firmware / model / docs subdirectories
- **License:** TBD (private project)

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## Changelog from v1

- **Removed rigid-flex solar tab** → small flat solar cell on back of card
- **Removed Qi wireless charging** (BQ51013B + 40×40 mm coil) — cost reduction
- **Removed NFC sticker** — replaced by dynamic e-paper QR code in demo mode
- **Replaced SPV1040 with BQ25570** — proper ultra-low-power harvester with brownout output
- **Replaced MCP73831 with MCP73832** — 500 mA fast charging instead of 150 mA
- **Fixed USB-C pin assignment:** GPIO19/20 (not GPIO43/44 — those are UART0)
- **Added USBLC6-2SC6** ESD protection on USB data lines
- **Upgraded e-paper:** 296×152 4-grayscale (GDEY029F51) instead of 296×128 B/W
- **Reduced touch pads from 5 to 3** (UP / SELECT / DOWN, vertical column)
- **Added MAX17048 fuel gauge** for accurate battery percentage reporting
- **Added 5 charge level LEDs** on back, driven by ESP32 GPIOs
- **Removed BOOT/RESET tactile buttons** → replaced by RST bridge pads (hot glue over after firmware load)
- **Larger LiPo:** 500 mAh (LP503035) instead of 300 mAh
- **Added 0.4 mm depth-controlled milling recess** for LiPo
- **SMD JST-PH connector** instead of through-hole — no through-hole soldering required
- **Pin reassignments** to avoid strapping pins (GPIO0/3/45/46) and PSRAM-internal pins (GPIO35/36/37)
- **Brownout protection:** BQ25570 VBAT_OK → XC6220 EN
- **Haptic motor moved to LiPo+** rail (not 3.3V) to prevent rail dipping